

AP Calc Lesson 2.9 Homework

Name _____

1. Which of the following equations is equivalent to $y = \sec x$? Select all that apply.

A) $y = \frac{1}{\sin x}$ E) $y = \sqrt{1 + \tan x}$

B) $y = \frac{1}{\sin x \cot x}$ F) $y = \sin x \tan x$

C) $y = \frac{\cos x}{1 - \sin^2 x}$ G) $y = \frac{\csc x}{\cot x}$

D) $y = \frac{\tan x}{\cos x}$ H) $y = \frac{\tan x + \cot x}{\csc x}$

2. Let $f(x) = \frac{\cot x + \cos x}{\cos x}$

a. Simplify $f(x)$.

b. Find $f'(x)$.

c. Write an equation for the line tangent to f at $x = \frac{\pi}{6}$.

3. Let $f(x) = \frac{g(x)}{\cot x}$. Find $f'(x)$ for each of the following.

a. $g(x) = 1$

b. $g(x) = x$

c. $g(x) = \csc x$

4. The function s models the length of the shadow cast by a flagpole with a height of 10 meters based on the angle of elevation of the sun. Let $s(\theta) = 10 \cot \theta$, where $s(\theta)$ is measured in meters and θ is the angle of elevation in radians.

a. Find $\frac{ds}{d\theta}$.

b. For $0 < \theta < \frac{\pi}{2}$, describe how s changes as θ increases.

c. Find the rate at which the length of the flagpole's shadow is changing when the angle of elevation of the sun is 1 radian. Indicate units of measure.

5. The table gives selected values of a differentiable function f .

x	$f(x)$	$f'(x)$
$\frac{\pi}{6}$	1	4
π	3	1.5

a. Let $g(x) = f(x) \cot x$. Find $g'\left(\frac{\pi}{6}\right)$.

b. Let $h(x) = \frac{f(x)}{\sec x - 1}$. Find $h'(\pi)$.

6. Let g be the function given by $g(x) = x + \tan x$.

a. Explain why a line tangent to g cannot be horizontal.

b. What is the slope of the curve at $x = \frac{5\pi}{4}$?

c. Find the value of x on the interval $[2, 4]$ such that the slope of the curve is 2.

7. The line tangent to the graph of $y = 3 \sec x$ at $x = \frac{\pi}{6}$ is parallel to the line $Ax - By = 10$. How many times greater is A than B ? Show the work that leads to your solution.

-  8. A line tangent to the graph of $y = \cot x$ at the point $(k, \cot k)$ also passes through the point $(2, 0)$. If $2 < k < \pi$, what is the value of k ?

9. Evaluate $\lim_{h \rightarrow 0} \frac{\csc\left(\frac{5\pi}{6} + h\right) - \csc\left(\frac{5\pi}{6}\right)}{h}$

- A) $-2\sqrt{3}$
- B) -2
- C) 2
- D) $2\sqrt{3}$
10. Let $g(x) = \cot x$. Which of the following gives the x -value at which the slope of the line tangent to g is -3 ?
- A) $f^{-1}(-3)$ where $f(x) = -\csc^2 x$
- B) $f(-3)$ where $f(x) = -\csc^2 x$
- C) $f^{-1}(-3)$ where $f(x) = \sec^2 x$
- D) $f(-3)$ where $f(x) = \sec^2 x$